# Annex V. Public Debt Sustainability Analysis

*The Netherlands have moderate public debt and low financing needs. Public debt is expected to remain sustainable, decreasing at a faster rate than previously anticipated from 57 percent of GDP in 2017 to 42 percent of GDP by 2023, owing to favorable growth developments. The largest risks to the baseline scenario are represented by a negative growth shock and a contingent liability shock, which could push up the debt-to-GDP ratio to 55 percent and 72 percent by 2020, respectively.*
The baseline scenario assumes a steady fiscal adjustment, albeit not in structural terms. It is underpinned by the following assumptions:

- *Real GDP growth* is forecast to stabilize around 2 percent over the projection period, driven by dynamic non-residential investment and strengthening private consumption supported by improving housing prices, while external demand would remain subdued. This is slightly higher than the medium-term growth potential.

- *Inflation* is projected to nominally increase to around 1.6 in 2019 and then stabilize at 1.4 percent over the medium run reflecting stronger domestic demand

- *Public debt* would decline toward 42 percent of GDP by the end of the projection period owing to strong revenue collection and contained expenditure, but remain above 50 percent until end-2019.

**Realism of Baseline Assumptions**

**1. The baseline projections are underpinned by realistic assumptions. Over 2007–15, staff projections of the main macroeconomic and fiscal variables in the Netherlands have constantly remained close to the median within the 25–75 interquartile range vis-à-vis other surveillance countries, except for inflation in 2008.** On average, real growth and primary balance forecasts appear to have been relatively conservative, while inflation forecasts have been slightly optimistic.

**2. Under the baseline scenario, public debt would rapidly decrease from 57 percent of GDP in 2017 to about 43 percent by the end of the projection period, under the joint effects of steady nominal growth and the pursuit of expenditure-based fiscal consolidation.** Public debt would increase up to 59 percent of GDP in the historical scenario, due to conservative growth and primary balance assumptions. It would steadily decrease over the medium term in all other scenarios.

**3. Under the baseline scenario, the primary balance would continue to decrease, reducing debt over time.** The contribution of the interest rate-growth differential will become more favorable, reflecting steady growth and very low effective interest rates, especially from 2018 onwards. The gross financing needs-to-GDP ratio is forecast to decline to less than zero percent of GDP by the end of the projection period.



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**Baseline Scenario and Stress Tests:**

**4.** The debt and financing needs projections are particularly sensitive to combined macro-fiscal, growth, and contingent liabilities shocks:

- **Growth shock.** Assuming a negative one standard deviation shock on the real growth rates in 2017–18, lowering them by about 2 ¼ percentage points compared to the baseline scenario, associated with inflation rates lower by 0.4 percentage points, public debt would increase to 55 percent of GDP in 2020 before gradually decreasing to 48 percent by 2023. Gross financing needs would peak at 3.1 percent of GDP in 2020 and remain at that level until the end of the projection period.

- **Primary balance shock.** A deterioration of the primary balance by 1 percentage point in 2017 and 0.8 percentage point in 2018 would lead to decreasing debt-to-GDP ratios to around 44 percent until the end of the projection period, with gross financing needs decreasing to 1.7 percent of GDP by 2023 after peaking at 1.9 in 2020.

- **Contingent liability shock.** A non-interest expenditure shock arising from the need to bail out 10 percent of the banking sector, accompanied with lower growth rates by one standard deviation, lower inflation, and higher interest rates, would push up gross financing needs to 20 percent of GDP in 2020 and public debt to 72 percent of GDP in 2020. Public debt would decrease somewhat to 65 percent of GDP by 2023 but gross financial needs would remain high through the end of the projection period.

**5. The fan charts show low uncertainty around the baseline.** The width of the symmetric fan chart, estimated at around 25 percent of GDP, illustrates the degree of uncertainty for equal probability upside and downside shocks. But given strong demand, upside shocks to growth and primary balance are unconstrained in the asymmetric fan chart, resulting in a downward sloping debt path.

**6. On average, the balance of risks is skewed to the upside with low debt weaknesses.** While external financing needs are well below warning benchmarks, there is a sizeable share of foreign liabilities that is above the upper warning threshold and constitutes a potential risk. However, overall debt levels are below the lower-risk assessment early warning benchmarks.

**7. The Netherlands's debt profile reveals some weaknesses.** While external financing needs are well below the lower risk-assessment early warning benchmark, the public debt held by nonresidents is above the upper benchmark. Spreads and the change in short-term debt are below the lower-risk assessment early warning benchmarks.



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## Figure A1. Netherlands: Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

(in percent of GDP unless otherwise indicated)

### Debt, Economic and Market Indicators [1/]

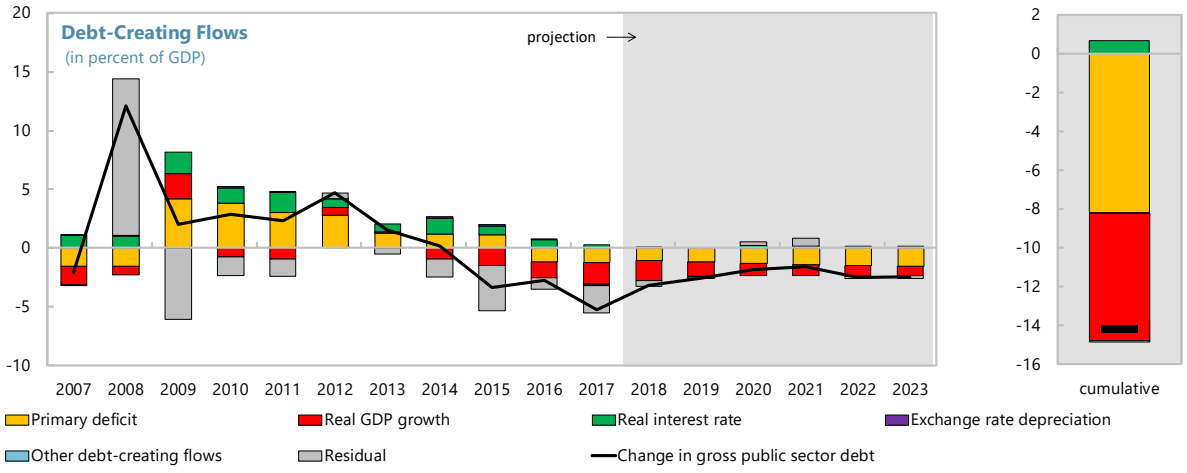
| | Actual | | | Projections | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2007-2015 [2/] | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
| Nominal gross public debt | 60.1 | 61.8 | 56.5 | 53.3 | 50.7 | 48.9 | 47.3 | 44.8 | 42.3 |
| Public gross financing needs | 2.8 | -0.4 | -0.5 | -0.5 | -0.7 | -0.8 | -0.9 | -1.0 | -1.0 |
| Public debt (in percent of potential GDP) | 59.7 | 61.2 | 56.5 | 53.9 | 51.6 | 49.8 | 48.3 | 45.7 | 43.2 |
| Real GDP growth (in percent) | 0.8 | 2.2 | 3.2 | 3.1 | 2.4 | 2.1 | 2.0 | 1.9 | 1.9 |
| Inflation (GDP deflator, in percent) | 1.1 | 0.6 | 1.1 | 1.2 | 1.6 | 1.2 | 1.4 | 1.4 | 1.4 |
| Nominal GDP growth (in percent) | 1.9 | 2.8 | 4.3 | 4.4 | 4.0 | 3.3 | 3.4 | 3.3 | 3.3 |
| Effective interest rate (in percent) [4/] | 3.2 | 1.7 | 1.6 | 1.4 | 1.5 | 1.6 | 1.6 | 1.7 | 1.8 |

**As of May 03, 2018**

| Sovereign Spreads | | |
|---|---|---|
| EMBIG (bp) 3/ | | 7 |
| 5Y CDS (bp) | | 26 |
| Ratings | Foreign | Local |
| Moody's | Aaa | Aaa |
| S&Ps | AA+ | AA+ |
| Fitch | AAA | AAA |

### Contribution to Changes in Public Debt

| | Actual | | | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2007-2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | cumulative | debt-stabilizing primary balance [9/] |
| Change in gross public sector debt | 2.2 | -2.8 | -5.3 | -3.2 | -2.6 | -1.9 | -1.6 | -2.5 | -2.5 | -14.2 | |
| Identified debt-creating flows | 2.4 | -1.8 | -3.0 | -2.7 | -2.5 | -2.2 | -2.3 | -2.3 | -2.2 | -14.1 | |
| Primary deficit | 1.6 | -1.2 | -1.3 | -1.1 | -1.2 | -1.3 | -1.4 | -1.5 | -1.6 | -8.2 | -0.7 |
| Primary (noninterest) revenue and grants | 42.7 | 43.6 | 43.5 | 43.5 | 43.5 | 43.5 | 43.5 | 43.5 | 43.5 | 261.2 | |
| Primary (noninterest) expenditure | 44.3 | 42.4 | 42.3 | 42.4 | 42.3 | 42.2 | 42.1 | 42.0 | 42.0 | 253.0 | |
| Automatic debt dynamics [5/] | 0.8 | -0.7 | -1.7 | -1.6 | -1.3 | -0.8 | -0.8 | -0.7 | -0.7 | -5.9 | |
| Interest rate/growth differential [6/] | 0.8 | -0.7 | -1.6 | -1.6 | -1.3 | -0.8 | -0.8 | -0.7 | -0.7 | -5.9 | |
| Of which: real interest rate | 1.1 | 0.7 | 0.3 | 0.1 | -0.1 | 0.2 | 0.1 | 0.1 | 0.2 | 0.6 | |
| Of which: real GDP growth | -0.4 | -1.4 | -1.9 | -1.7 | -1.2 | -1.0 | -0.9 | -0.9 | -0.8 | -6.6 | |
| Exchange rate depreciation [7/] | 0.0 | 0.0 | -0.1 | ... | ... | ... | ... | ... | ... | ... | |
| Other identified debt-creating flows | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| Please specify (1) (e.g., drawdown of deposits) | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| Contingent liabilities | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| Please specify (2) (e.g., ESM and Euroarea loans) | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| Residual, including asset changes [8/] | -0.1 | -0.9 | -2.3 | -0.5 | -0.1 | 0.3 | 0.7 | -0.2 | -0.2 | -0.1 | |




Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

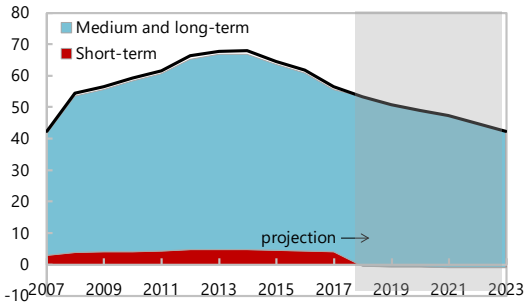


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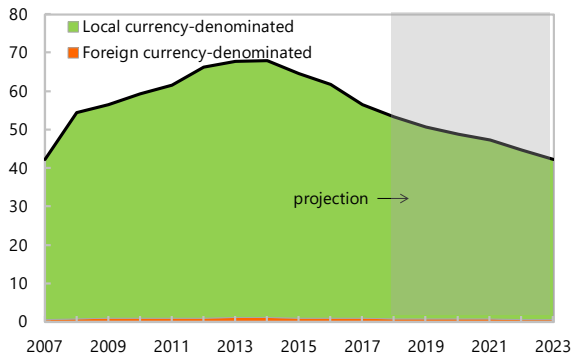
## Figure A2. Netherlands: Public DSA—Composition of Public Debt and Alternative Scenarios

### Composition of Public Debt

**By Maturity**
(in percent of GDP)



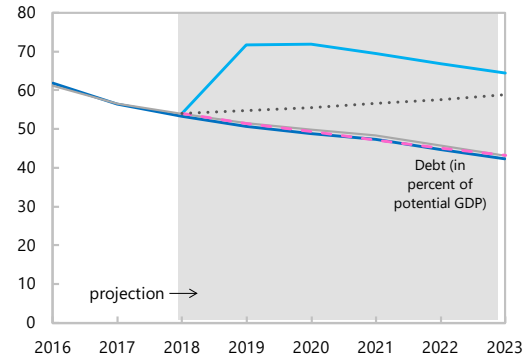

**By Currency**
(in percent of GDP)



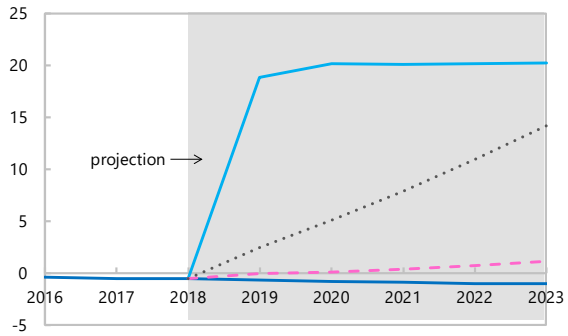

### Alternative Scenarios

Baseline
Contingent Liability Shock
Historical
Constant Primary Balance

**Gross Nominal Public Debt**
(in percent of GDP)




**Public Gross Financing Needs**
(in percent of GDP)




### Underlying Assumptions
(in percent)

| Baseline Scenario | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|
| Real GDP growth | 3.1 | 2.4 | 2.1 | 2.0 | 1.9 | 1.9 |
| Inflation | 1.2 | 1.6 | 1.2 | 1.4 | 1.4 | 1.4 |
| Primary Balance | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 1.6 |
| Effective interest rate | 1.4 | 1.5 | 1.6 | 1.6 | 1.7 | 1.8 |
| **Constant Primary Balance Scenario** | | | | | | |
| Real GDP growth | 3.1 | 2.4 | 2.1 | 2.0 | 1.9 | 1.9 |
| Inflation | 1.2 | 1.6 | 1.2 | 1.4 | 1.4 | 1.4 |
| Primary Balance | 1.1 | 1.1 | 1.1 | 1.1 | 1.1 | 1.1 |
| Effective interest rate | 1.4 | 1.5 | 1.6 | 1.7 | 1.8 | 1.9 |

| Historical Scenario | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|
| Real GDP growth | 3.1 | 0.9 | 0.9 | 0.9 | 0.9 | 0.9 |
| Inflation | 1.2 | 1.6 | 1.2 | 1.4 | 1.4 | 1.4 |
| Primary Balance | 1.1 | -1.3 | -1.3 | -1.3 | -1.3 | -1.3 |
| Effective interest rate | 1.4 | 1.5 | 1.8 | 2.0 | 2.2 | 2.5 |
| **Contingent Liability Shock** | | | | | | |
| Real GDP growth | 3.1 | 0.4 | 0.0 | 2.0 | 1.9 | 1.9 |
| Inflation | 1.2 | 1.1 | 0.7 | 1.4 | 1.4 | 1.4 |
| Primary Balance | 1.1 | -17.7 | 1.3 | 1.4 | 1.5 | 1.6 |
| Effective interest rate | 1.4 | 1.7 | 3.3 | 2.1 | 2.3 | 2.4 |

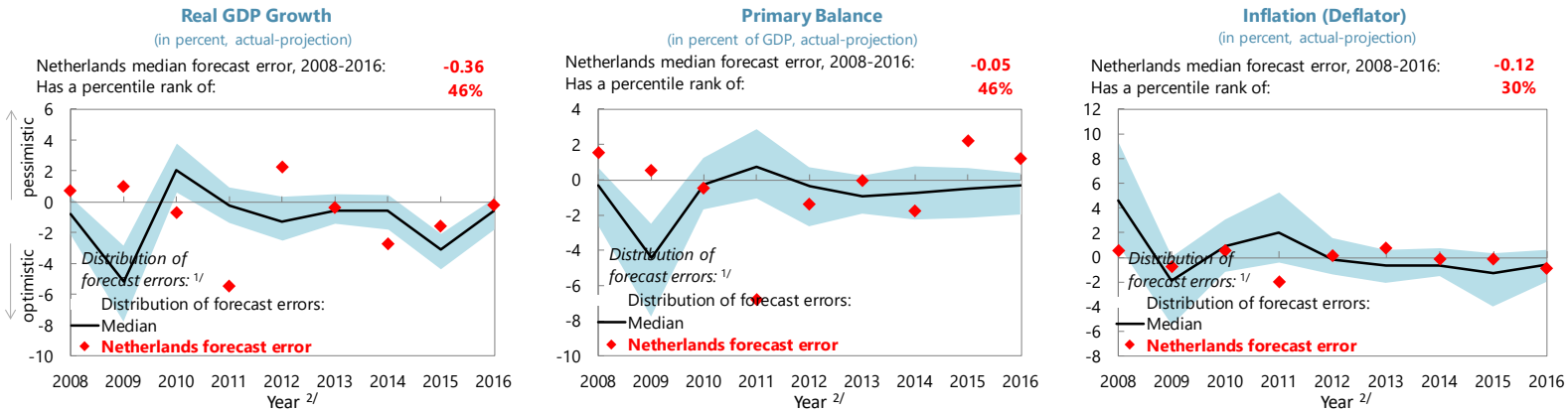
Source: IMF staff.



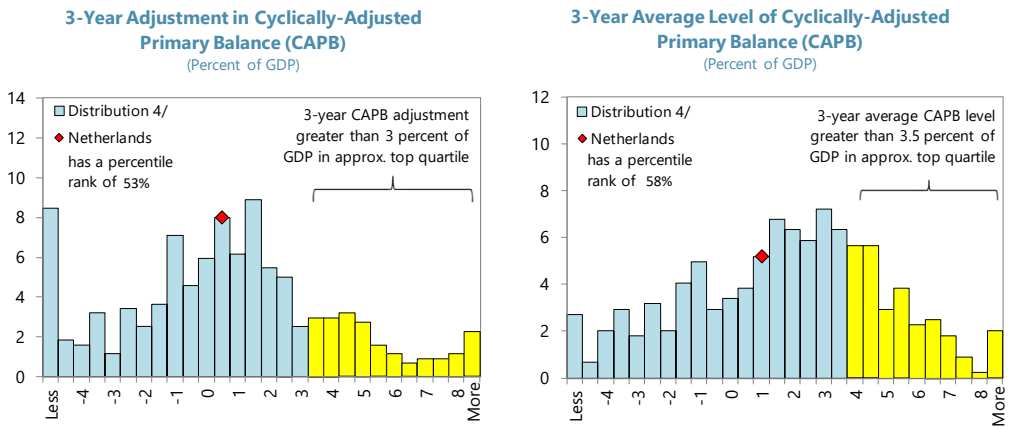
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## Figure A3. Netherlands: DSA—Realism of Baseline Assumptions

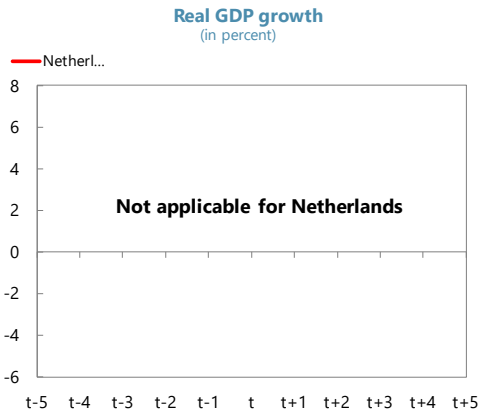
### Forecast Track Record, versus surveillance countries




### Assessing the Realism of Projected Fiscal Adjustment




### Boom-Bust Analysis 3/




Source : IMF Staff.

1/ Plotted distribution includes surveillance countries, percentile rank refers to all countries.

2/ Projections made in the spring WEO vintage of the preceding year.

3/ Not applicable for Netherlands, as it meets neither the positive output gap criterion nor the private credit growth criterion.

4/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.

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